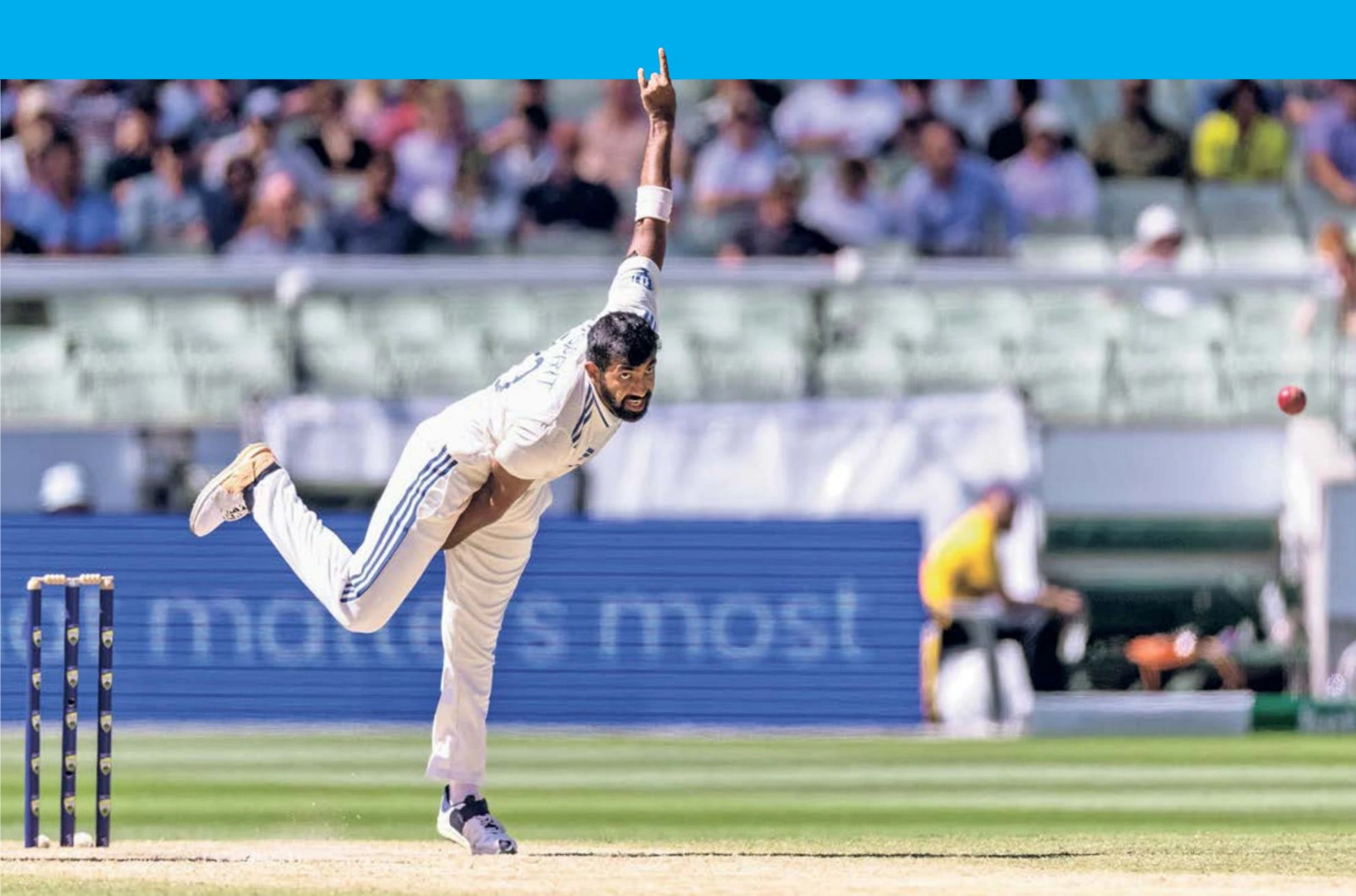




represents the most severe end of what's known as the bone stress injury (BSI) continuum," Leipus explains. "Put simply, bone is a living, adaptable tissue that responds to mechanical load. When the stress placed on it consistently exceeds its ability to recover and adapt, the breakdown process outpaces repair, eventually leading to a cortical fracture."

Bowling pace is inherently risky; not every injury is avoidable, and sometimes even well-managed workloads can lead to breakdowns. But in most cases, patterns of mismanagement are clear and avoidable.

Leipus says that the progression of a BSI to a fracture in a pacer is rarely caused by a single factor. "They're typically the result of a complex interplay between biomechanics, workload errors, and systemic gaps in preparation or monitoring," he explains. Add to it: the player's age, bowling experience, various physical attributes, and individual biomechanical issues.

Ian Pont, head coach and founder of the National Fast Bowling Academy in the United Kingdom, doesn't find a problem in Bumrah's action, which he describes as "unique and biomechanically efficient."

"He did suffer injuries, but they weren't due to his action – more due to workload mismanagement; excessive play without rest," Pont, who has coached in India for over 15 years, says.

Rest and recovery are crucial aspects of a fast bowler's longevity. If the pacer is tired, fatigued, or sore, physiotherapists usually take it as a red flag for a stress reaction. Workload management is the science of planning how many balls a bowler delivers, how often, and at what intensity.

"It is the key to keeping bowlers injury-free and performing well over time. Done right, it extends careers and prevents stress fractures. Done poorly, it leads to injury, rehab, and re-injury cycles," Pont says.

According to Pont, the support staff – bowling coach,

'Back' at it: Jasprit Bumrah, the world's top-ranked Test bowler, injured his back during the fifth Test against Australia in early January and was forced to miss England's white-ball tour to India and the Champions Trophy. GETTYIMAGES

physiotherapist, and strength and conditioning coach – should take note of each delivery a pacer bowls, during a game and while training. "Increase weekly workload gradually," he says. "No more than a 20 per cent jump week to week. For example, going from 60 to 100 balls weekly too quickly raises injury risk."

Body physiology and mechanics

Bumrah's case isn't an isolated one. A decade ago, a young Indian speedster too found himself in a loop of back injuries.

By 2012, promising fast bowler Varun Aaron had developed a recurring back injury that had already ruled him out of India's 2011 tour to Australia and limited his